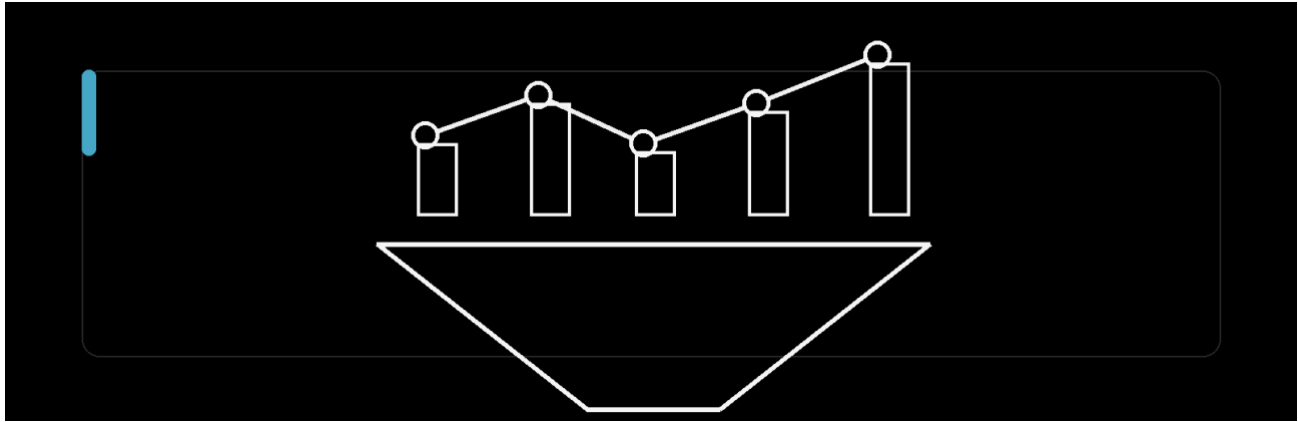




Funnel Analysis

Tags	EDA · Excel · Python · Funnel Analysis · Statistical Testing
Description	Customer Journey Optimisation and Resource Prioritisation

Executive Summary

The analysis covers **12,719 customer-journey events** organised into **5,000 sessions**. A total of **1,010 sessions** reached purchase, producing an overall session purchase rate of **20.2%**.

The main break occurs during product evaluation. Of the 3,987 sessions that reached a product page, only 1,599 progressed to the cart. The resulting **59.9% drop-off** removes **2,388 sessions** at one transition, making product page-to-cart progression the clearest improvement opportunity.

Device, represented-region, referral-source and time-of-day purchase rates are close to one another. Pearson chi-square tests do not show statistically reliable segment differences, so the current evidence supports improving the shared product experience before pursuing a broad segment-specific redesign.

Problem

Traffic reaches the site, but most sessions do not progress to purchase. The analysis therefore asks where customers disengage, which patterns are supported by the event data, and where product and analytics resources are most likely to produce measurable value.

Objective

Reconstruct the journey at session level, quantify retention and loss between stages, compare outcomes across major segments, and translate the findings into an evidence-led optimisation and experimentation plan.

Analytical Approach

The source data is event-level: each row records a session interaction such as a home visit, product-page view, cart visit or checkout visit. The analysis first reorganised these rows into complete session journeys and then measured how many sessions reached each successive stage.

How the Data Was Prepared

- **Reconstructed journeys.** Events were ordered by timestamp within each session to recover the path from entry to purchase.
- **Validated funnel logic.** Stage order, repeated stages, duplicate rows, missing values and within-session attribute consistency were checked.
- **Aggregated at session level.** Headline funnel and segment rates use distinct sessions rather than raw event rows.
- **Compared key segments.** Purchase outcomes were compared by device, represented region, referral source and six-hour time window.
- **Separated evidence from hypotheses.** The funnel identifies where friction occurs; possible causes still require instrumentation or controlled tests.

Dataset and Unit of Analysis

Measure	Value	How it is used
Event records	12,719	Raw interactions used to reconstruct journeys
Sessions	5,000	Primary denominator for funnel and segment rates
Unique users	1,872	Shows that some users generated multiple sessions
Purchase sessions	1,010	Sessions with a purchase outcome
Observation period	January–June 2025	Timestamp range represented in the source data

Data-quality result

No missing values, duplicate event rows, repeated funnel stages, stage-order problems or within-session device, country and referral conflicts were found. The two uploaded workbooks contain the same 12,719-row event table.

Technical Implementation

Python was used to validate the event data, order interactions, rebuild sessions, calculate funnel metrics and run significance tests. Excel supported pivot-table checks and visual review. The workflow is repeatable, but the source workbooks do not include timing logs, so no percentage time-saving is claimed.

Funnel Overview

The funnel measures how many sessions remain at each stage. Because the stages are sequential, two measures matter: the absolute number of sessions retained and the conversion rate from the immediately preceding stage.

Sessions retained at each stage



Figure 1. Sessions retained at each stage of the customer journey.

Stage	Sessions	Stage conversion	Drop-off	Sessions lost
Home	5,000	—	—	—
Product page	3,987	79.7%	20.3%	1,013
Cart	1,599	40.1%	59.9%	2,388
Checkout	1,123	70.2%	29.8%	476
Purchase	1,010	89.9%	10.1%	113

What the Funnel Shows

Entry into the product experience is comparatively healthy: 79.7% of home sessions reach a product page. The major loss comes next. Only 40.1% of product-page sessions continue to the cart, removing 2,388 sessions from the journey at a single transition.

Progression improves after cart entry. About 70.2% of cart sessions reach checkout and 89.9% of checkout sessions reach purchase. Later stages still deserve monitoring, but the largest available gain sits earlier, while customers are deciding whether an item is worth adding.

Business interpretation

The bottleneck is not simply “low overall conversion.” It is specifically weak product page-to-cart progression. That sharper diagnosis gives product, design and analytics teams a concrete metric to improve: product-to-cart conversion.

Product Evaluation Bottleneck

The product page-to-cart transition is both the largest percentage loss and the largest absolute loss. Improving this stage can affect more sessions than an equivalent percentage improvement later in the funnel, which makes it the strongest candidate for initial resource allocation.

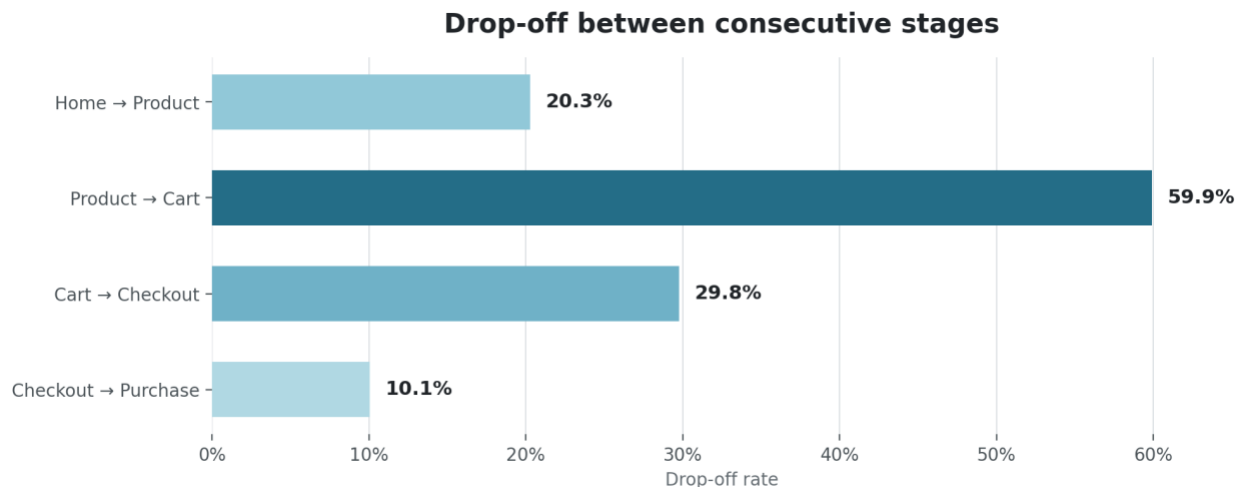


Figure 2. Drop-off rates between consecutive funnel stages.

What the Data Supports

- **The location of friction is clear.** Most loss occurs after a product page is reached but before an item is added to the cart.
- **The loss is large in both relative and absolute terms.** The transition drops 59.9% of eligible sessions, equal to 2,388 sessions.
- **Later-stage completion is comparatively strong.** Once a session reaches checkout, roughly nine out of ten proceed to purchase.
- **The pattern is shared.** Device and represented-region rates are nearly identical, so the evidence does not point to one isolated segment.

Potential Causes to Test

Hypothesis	Controlled test	Primary metric
Pricing clarity	Show total cost, delivery terms and key conditions earlier	Product → cart rate
Value and reassurance	Strengthen benefits, reviews, returns and trust messaging	Product → cart rate
Product information	Test richer imagery, specifications, comparisons and FAQs	Product → cart rate
CTA usability	Test wording, prominence, placement and interaction feedback	Cart entry rate
Variant or stock friction	Instrument selection errors and unavailable combinations	Successful cart additions

Evidence boundary

The event data records behaviour, not motivation. Pricing, value, trust, information quality and interface design are plausible explanations, but none should be presented as the proven cause until additional behavioural signals or controlled experiments support the claim.

Referral Source Analysis

Referral analysis answers two different questions: which source contributes the greatest number of purchases, and which source converts its sessions most efficiently. Purchase share reflects volume as well as performance, so it should not be treated as a conversion metric on its own.

Referral source	Sessions	Purchases	Purchase rate	Share of purchases
Google	1,280	277	21.64%	27.43%
Email	1,251	251	20.06%	24.85%
Direct	1,226	243	19.82%	24.06%
Social Media	1,243	239	19.23%	23.66%

Interpretation

Google contributes the largest number of purchases at 277 and has the highest descriptive purchase rate at 21.64%. Social Media has the lowest descriptive rate at 19.23%. The difference between the highest and lowest rates is only 2.41 percentage points.

A Pearson chi-square test of independence does not show a statistically reliable relationship between referral source and purchase outcome ($\chi^2(3) = 2.50$, $p = .475$, Cramér's $V = .022$). The observed ranking is useful for monitoring and hypothesis generation, but the effect is very small and is not strong enough to justify a major budget shift by itself.

Resource Implication

- **Do not increase acquisition spend solely because one source produced more purchases.** The source may simply have generated more sessions.
- **Improve the shared product-page experience first.** All referral sources feed into the same large product-to-cart loss.
- **Evaluate channels with cost and value measures.** Add customer acquisition cost, revenue per session and customer value before reallocating budget.
- **Use channel-specific experiments after the product baseline improves.** Landing-page alignment and campaign messaging can then be tested with cleaner measurement.

Decision supported by the data

Prioritise conversion efficiency before buying more traffic. Improving a shared bottleneck can benefit every acquisition source, whereas additional traffic would send more sessions into the same weak transition.

Device and Geographic Patterns

Segment analysis checks whether the overall funnel problem is concentrated in a particular device or represented region. Here, the differences are extremely small. That changes the recommendation: strengthen the common experience before committing to a broad device-only or region-only redesign.

Device Comparison

Device	Sessions	Purchases	Purchase rate
Desktop	1,666	339	20.35%
Mobile	1,671	337	20.17%
Tablet	1,663	334	20.08%

Desktop, Mobile and Tablet purchase rates range from 20.08% to 20.35%, a spread of only 0.27 percentage points. The Pearson chi-square test is not significant ($\chi^2(2) = 0.04$, $p = .981$, Cramér's $V = .003$). This dataset therefore does not support describing mobile as the primary source of funnel friction.

Represented Regions

Region	Countries represented	Sessions	Purchases	Purchase rate
Europe	France, UK, Germany	2,194	447	20.37%
Americas	USA, Canada	1,421	287	20.20%
Asia-Pacific	India, Australia	1,385	276	19.93%

Europe, the Americas and Asia-Pacific all convert at approximately 20%. The comparison is not statistically reliable ($\chi^2(2) = 0.10$, $p = .949$, Cramér's $V = .005$). The uploaded data contains no African countries, so the report does not make a claim about Africa.

Practical conclusion

The near-identical rates suggest a structural experience issue rather than one segment performing unusually poorly. Continue measuring device and country results, but focus initial product work on the common product page-to-cart journey.

Time-of-Day Analysis

Session start times were grouped into four six-hour periods: late night (00:00–05:59), morning (06:00–11:59), afternoon (12:00–17:59) and evening (18:00–23:59). The comparison tests whether purchase outcomes vary across these time windows.

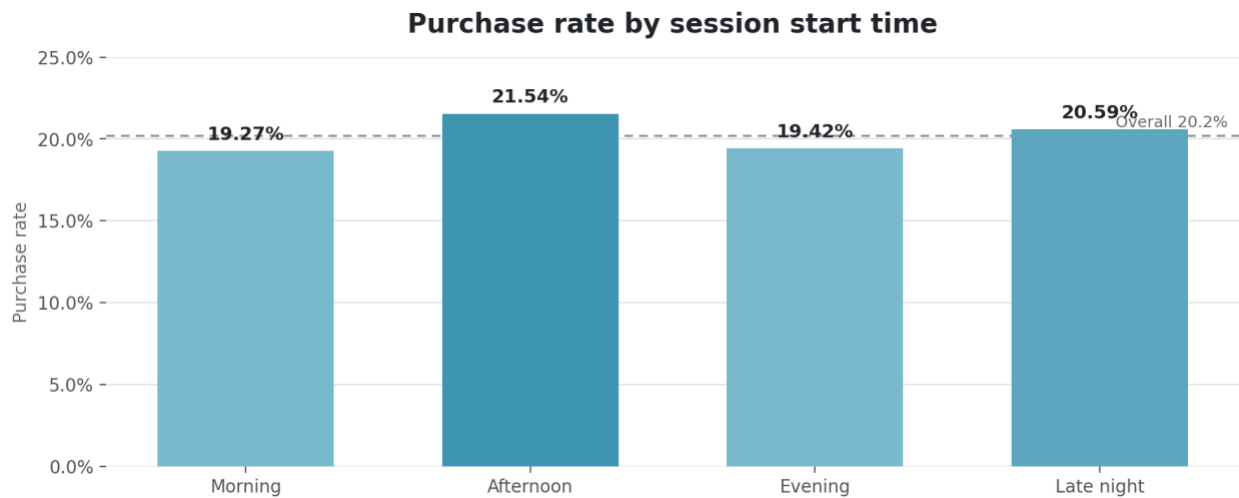


Figure 3. Purchase rate by session start time.

Time segment	Window	Sessions	Purchases	Purchase rate
Morning	06:00–11:59	1,261	243	19.27%
Afternoon	12:00–17:59	1,272	274	21.54%
Evening	18:00–23:59	1,277	248	19.42%
Late night	00:00–05:59	1,190	245	20.59%

Statistical Validation: Pearson Chi-Square

The test compares a 4×2 contingency table: four time segments by two outcomes (purchase and no purchase). The null hypothesis is that purchase outcome is independent of session start time. The result is $\chi^2(3) = 2.69$, $p = .442$, with Cramér's $V = .023$.

Because p is greater than .05, the observed differences are not statistically significant. Afternoon records the highest descriptive rate at 21.54%, while morning records 19.27%, but a difference of this size could reasonably arise from sampling variation. The very small Cramér's V also indicates little practical separation between the groups.

How to interpret the result

The chart can be used to propose an experiment, not to declare a permanent scheduling rule. A defensible recommendation is to test afternoon timing against another window using comparable audiences and a pre-defined primary metric.

Campaign Measurement Plan

- Randomise timing across comparable audiences instead of comparing naturally different traffic periods.
- Measure incremental purchase rate and revenue per session, not only opens or notification clicks.
- Track opt-outs, complaints, repeat engagement and customer value as guardrails.
- Retain the current schedule if the experiment does not produce a reliable and practically meaningful lift.

Recommendations and Resource Prioritisation

The recommendations below are ranked by the strength of the evidence and the size of the affected funnel stage. The aim is to move from diagnosis to measurable action without presenting untested explanations as established facts.

Priority	Workstream	Evidence strength	Why it comes first
1	Product-page experiments	High	59.9% product-to-cart drop-off and 2,388 lost sessions
2	Cart progression review	Medium	29.8% cart-to-checkout drop-off
3	Measurement and monitoring	High	Current analysis is retrospective
4	Referral and timing tests	Exploratory	Small, non-significant descriptive differences
5	Device or region redesigns	Low	Purchase rates are nearly identical

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Optimise product pages

Make total cost, delivery information, product benefits and reassurance easy to find. Test pricing clarity, product information, trust messaging and CTA presentation separately so the team can learn what actually changes cart entry.

2

Review cart-entry and cart progression

Verify that the add-to-cart interaction is prominent, responsive and accessible. Instrument variant-selection errors, stock problems, unexpected cost exposure and unclear next steps before simplifying the cart journey.

3

Build stage-level monitoring

Track product-to-cart rate as the primary leading metric, alongside absolute session losses, checkout completion, purchase rate, revenue per session and technical performance. Add device, country and referral filters for diagnosis rather than assumption.

4

Validate decisions with controlled tests

Predefine the primary metric, guardrails, sample-size requirement and decision threshold. Use A/B testing or carefully phased experiments rather than relying only on before-and-after comparisons.

Measurement rule

A product-to-cart improvement is valuable only if it also protects later-stage completion, revenue quality and customer experience. Report both stage movement and final business outcomes before adopting a variation.

Key Outcomes, Limitations and Next Steps

Key Outcomes

- Located the primary funnel bottleneck at the product evaluation stage.
- Quantified the bottleneck as a 59.9% drop-off and 2,388 sessions lost before cart entry.
- Showed that device and represented-region purchase rates are nearly identical.
- Separated descriptive referral and timing differences from statistically reliable effects.
- Produced a repeatable, session-level analysis workflow for future customer-journey data.

Limitations

The dataset does not include product identity, price, discount exposure, delivery cost, stock status, error logs, experiment assignment, revenue, marketing cost or qualitative feedback. It therefore supports behavioural diagnosis and test prioritisation, but it cannot directly establish why customers leave or estimate return on investment.

The analysis uses sessions as the primary unit because the funnel describes individual visits. The source contains 1,872 unique users, so future work could add user-level repeat behaviour, first-time versus returning status and longer-term customer value.

Next Analytical Steps

- Add product, price, stock, discount, delivery-cost, CTA-click and error-event fields to the event model.
- Create a monitored funnel dashboard with stage rates, absolute losses, segment filters and anomaly alerts.
- Launch the highest-priority product-page experiment and report both stage-level and final purchase outcomes.
- Add revenue, acquisition cost and customer-value measures before changing channel budgets.
- Maintain an experiment log containing the hypothesis, design, decision threshold, result and follow-up action.

Final decision

The strongest current action is to improve product evaluation and measure product-to-cart movement. Referral, timing, device and regional patterns should remain visible in monitoring, but the data does not support treating any one of them as the primary cause of low purchase progression.

Statistical Test Summary

Comparison	χ^2	df	p-value	Cramér's V	Interpretation
Device	0.04	2	.981	.003	No evidence of a device difference
Represented region	0.10	2	.949	.005	No evidence of regional underperformance
Referral source	2.50	3	.475	.022	Directional ranking only
Time of day	2.69	3	.442	.023	Test timing experimentally